New video to a source-bound scene candidate

Hashes bind source declarations; geometric fidelity and robot task success require separate validation

(a) Host: bind the inputs

Source video
Original bytes + source file hash

Observation manifest
Frames + nominal sampling times

Reconstruction stage
Raw video: cloud submit / job resume
or source-declared artifact manifest
PLY + hashes + receipt

worker_inputs
Bound PLY + scene spec + source record

(b) Worker: construct the scene

Visible Gaussian appearance
Identity or declared Sim(3) transform

Bounded voxel collision proxies
Estimated occupancy; no surface truth

USD layers + preparation record
Objects, transforms, provenance, limits

Optional explicit scene specification
Declared floor; box + prismatic button
Scale evidence; camera + render settings

(c) Independently inspect + deliver

Independent structural audit
Read exported USD + receipts anew
Layers, joints, finite bounds, bindings

Native camera survey
Camera path + frames + encoded video
Skipped by --prepare-only

Delivery receipt
CPU: SCENE_EXPORTED
Survey: SCENE_SURVEY_READY
Artifact hashes + actual verification scope

Robot task validated: false

FAILURE / RESUME Stop + retain evidence — Reuse only matching run identity + artifact hashes

No scale anchor: unverified — Unknown up: no floor guessed — Structural pass is not task success